

Control System: Phase 2

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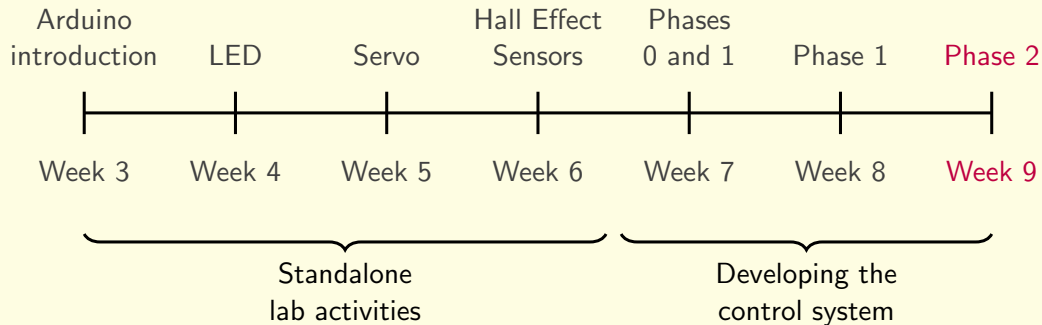
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MAC 233: **25/11/2026**



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MAC 233: LABS OVERVIEW



Everything is built and tested. Today: how good is the example controller, and how could you make it better?

Not finished step 5 yet? Do that first.

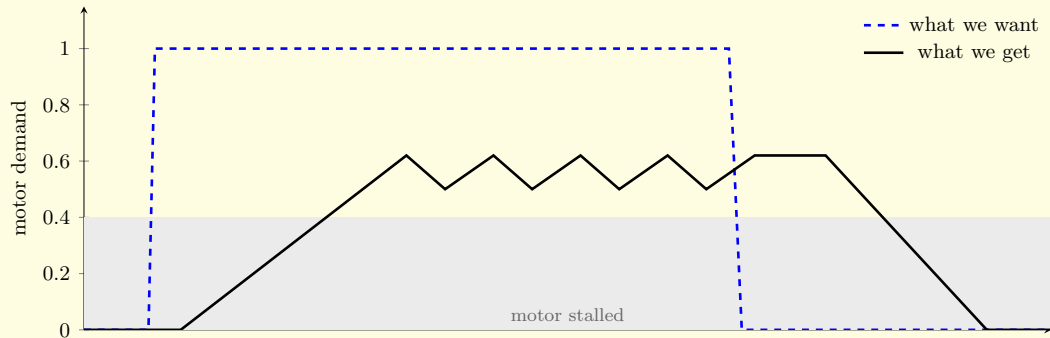
TESTING THE EXAMPLE CONTROLLER

Predict: sketch the motor demand and the wheel speed against time.

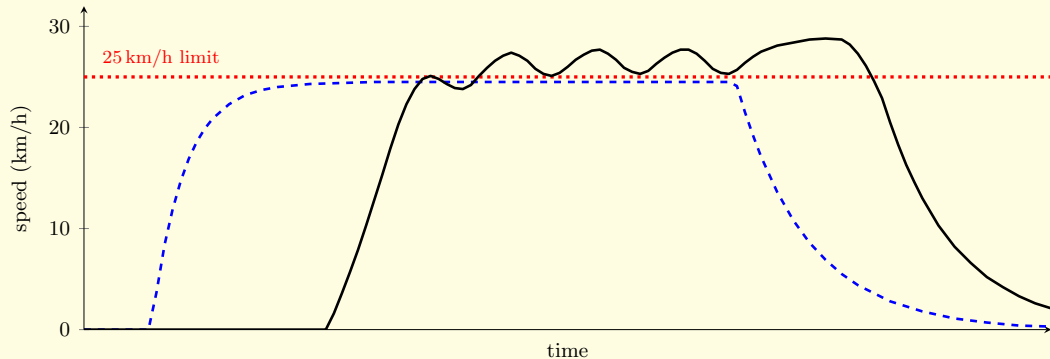
Test: turn the crank about once a second for 20 seconds, then stop.

Compare: copy the serial monitor output and plot it. Does it match your sketch?

WHAT WE WANT VS WHAT WE GET: MOTOR DEMAND



WHAT WE WANT VS WHAT WE GET: SPEED



Dashed: what we want. Solid: what we get.

THREE PROBLEMS

Slow start: the motor stalls until the demand reaches about 0.4.

Overshoot: the speed wobbles around the 25 km/h limit.

Late stop: the help carries on for up to 3 seconds after the rider stops.

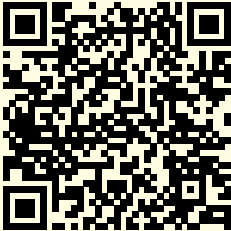
YOUR TURN

Change anything between the two lines of slashes in 'loop'. Leave the DO NOT CHANGE block alone.

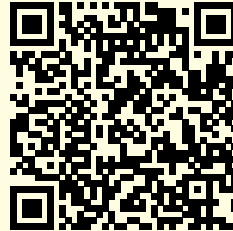
- ▶ Start the ramp from where your motor starts to turn.
- ▶ Ease off as the bike gets close to 25 km/h.
- ▶ Change the ramp rates.

Same method as Phase 1: one change at a time, predict, test, compare.

Worksheet



Sketch



Problems? Raise your hand and someone will come and help you.